

Expt 4 Design of Low Pass Filter using Windowing Techniques

4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique in MATLAB

Objective:

1. To design an FIR Low-Pass Filter using the Hamming Window technique for the given cutoff frequency, sampling frequency, and filter order specifications.
2. To generate the FIR filter coefficients using the Hamming Window method in MATLAB and implement the designed filter.
3. To obtain and analyze the magnitude response and phase response of the FIR Low-Pass Filter designed using the Hamming Window.

Procedure:

1. Open MATLAB and create a new script file.
2. Clear the workspace, command window, and close all figure windows.
3. Specify the sampling frequency, cutoff frequency, and filter order.
4. Calculate the normalized cutoff frequency.
5. Design the FIR Low-Pass Filter using the `fir1()` function with the Hamming Window (`hamming`).
6. Generate and display the FIR filter coefficients.
7. Obtain the magnitude and phase responses of the designed filter using the `freqz()` function.
8. Plot and analyze the frequency response characteristics of the filter.
9. Verify the low-pass filtering behavior by observing the passband and stopband responses.
10. Record the observations and conclude the performance of the FIR Low-Pass Filter designed using the Hamming Window technique.

Program (Matlab Code):

```
clc;  
  
clear;  
  
close all;
```

```
%% --- Input Parameters ---
```

```
fp = 1000;      % Passband cutoff frequency (Hz)
```

```
fs = 4000;      % Sampling frequency (Hz)
```

```
N = 50;         % Filter order
```

```
wc = fp/(fs/2); % Normalized cutoff frequency
```

```
%% --- FIR Low Pass using Hamming Window ---
```

```
b_hamm = fir1(N, wc, 'low', hamming(N+1));
```

```
%% --- Display Hamming Window Coefficients ---
```

```
disp('--- Hamming Window Coefficients ---');
```

```
disp(b_hamm');
```

```
%% --- Plot Frequency Response ---
```

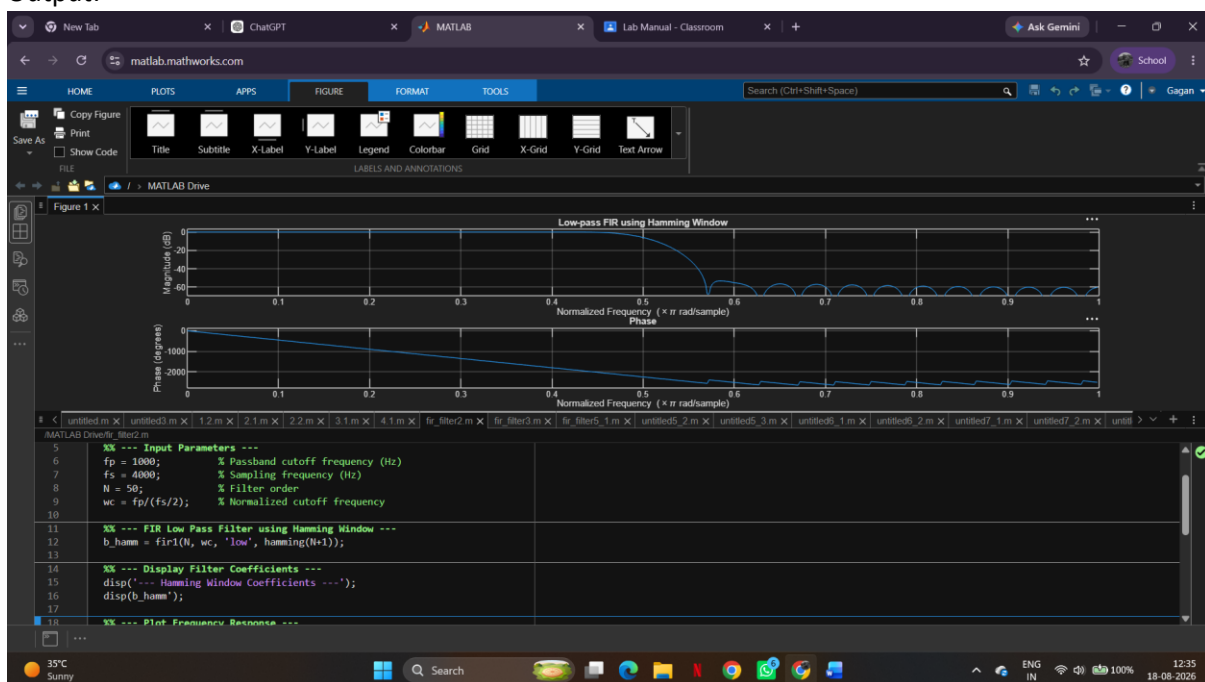
```
figure;
```

```
freqz(b_hamm, 1);
```

```
title('Low-pass FIR using Hamming Window');
```

```
grid on;
```

Output:



--- Hamming Window Coefficients ---

0.0010
-0.0000
-0.0013
0.0000
0.0021
-0.0000
-0.0034
0.0000
0.0055
-0.0000
-0.0084
0.0000
0.0125
-0.0000
-0.0181
0.0000
0.0260
-0.0000
-0.0379
0.0000
0.0580
-0.0000
-0.1026
0.0000
0.3168
0.4995
0.3168
0.0000
-0.1026
-0.0000

0.0580

0.0000

-0.0379

-0.0000

0.0260

0.0000

-0.0181

-0.0000

0.0125

0.0000

-0.0084

-0.0000

0.0055

0.0000

-0.0034

-0.0000

0.0021

0.0000

-0.0013

-0.0000

0.0010

Result:

1. The FIR Low-Pass Filter was successfully designed using the Hamming Window technique for the specified filter parameters.
2. The FIR filter coefficients were generated and implemented successfully in MATLAB.
3. The magnitude and phase responses of the designed filter were obtained and analyzed.
4. The filter exhibited the desired low-pass characteristics with improved stopband attenuation compared to the Rectangular Window method.